

Student ID: **DAYYAM JEEVANA REDDY** [193A11025882](#)

3 / 3 Submitted

Q1. Smart Hospital Patient Registration System

Score: 4.00 TC: 6/6 passed

CODE

```

import java.util.*;

public class Patient {
    private String patientId;
    private String name;
    private String mobileNumber;
    private String department;
    private String doctorName;

    public Patient(String patientId, String name, String mobileNumber, String department, String doctorName) {
        this.patientId = patientId;
        this.name = name;
        this.mobileNumber = mobileNumber;
        this.department = department;
        this.doctorName = doctorName;
    }

    public String getPatientId() { return patientId; }
    public String getName() { return name; }
    public String getMobileNumber() { return mobileNumber; }
    public String getDepartment() { return department; }
    public String getDoctorName() { return doctorName; }

    private static List<Patient> patientList = new ArrayList<>();
    private static Map<String, Patient> patientMap = new HashMap<>();
    private static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {
        while (scanner.hasNextLine()) {
            String inputLine = scanner.nextLine().trim();
            if (inputLine.isEmpty()) continue;

            int choice;
            try {
                choice = Integer.parseInt(inputLine);
            } catch (NumberFormatException e) {
                continue;
            }

            switch (choice) {
                case 1:
                    registerPatient();
                    break;
                case 2:
                    searchPatient();
                    break;
                case 3:
                    displayAllPatients();
                    break;
                case 4:
                    countDepartmentList();
                    break;
                case 5:
                    return;
            }
        }
    }

    private static void registerPatient() {
        if (scanner.hasNextLine()) return;
        String id = scanner.nextLine().trim();
        if (scanner.hasNextLine()) return;
        String name = scanner.nextLine().trim();
        if (scanner.hasNextLine()) return;
        String mobile = scanner.nextLine().trim();

```

```

        if (!scanner.hasNextLine()) return;
        String dept = scanner.nextLine().trim().toLowerCase();
        if (!scanner.hasNextLine()) return;
        String doc = scanner.nextLine().trim();
        if (patientMap.containsKey(id)) return;
        if (name.matches("[a-zA-Z\\s]+")) return;
        if (email.matches("\\d{10}")) return;
        name = toUpperCase(name);
        Patient newPatient = new Patient(id, name, email, dept, doc);
        patientList.add(newPatient);
        patientMap.put(id, newPatient);
        System.out.println("Patient Registered Successfully.");
    }

    private static void searchPatient() {
        if (!scanner.hasNextLine()) return;
        String id = scanner.nextLine().trim();
        if (patientMap.containsKey(id)) {
            Patient p = patientMap.get(id);
            System.out.println("Patient Details");
            System.out.println("ID : " + p.getId());
            System.out.println("Name : " + p.getName());
            System.out.println("Email : " + p.getEmail());
            System.out.println("Department : " + p.getDepartment());
            System.out.println("Doctor : " + p.getDoctorName());
        } else {
            System.out.println("Patient Not found");
        }
    }

    private static void displayAllPatients() {
        if (patientList.isEmpty()) return;
        System.out.println(".....");
        System.out.println("All have important");
        System.out.println(".....");
        for (Patient p : patientList) {
            System.out.println(p.getId() + " " + p.getName() + " " + p.getDepartment());
        }
    }

    private static void countDepartment() {
        if (patientList.isEmpty()) return;
        Map<String, Integer> deptCountMap = new HashMap<>();
        for (Patient p : patientList) {
            String dept = p.getDepartment();
            deptCountMap.put(dept, deptCountMap.getOrDefault(dept, 0) + 1);
        }
        System.out.println("Department Count");
        for (Map.Entry<String, Integer> entry : deptCountMap.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }

    private static String toUpperCase(String input) {
        if (input == null || input.isEmpty()) return input;
        String[] words = input.split("\\s+");
        StringBuilder properCase = new StringBuilder();
        for (String word : words) {
            if (!word.isEmpty()) {
                properCase.append(Character.toUpperCase(word.charAt(0)))
                    .append(word.substring(1).toLowerCase())
                    .append(" ");
            }
        }
        return properCase.toString().trim();
    }
}

```

INPUT

```
1
P001
Rahul Kumar
9876543210
CardNo2000
Dr. Arun
1
P002
Priya
9876543211
Dr. Isha
Dr. Arun
1
2
P001
0
```

OUTPUT

```
Patient Registered Successfully.
Patient Registered Successfully.
.....
To view Department
.....
P001 Rahul Kumar CardNo2000
P002 Priya Dr. Isha
Department Count
CardNo2000 : 1
Dr. Isha : 1
Patient Details
Id : P001
Name : Rahul Kumar
Mobile : 9876543210
Department : CardNo2000
Doctor : Dr. Arun
```

```

import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        ArrayList<Book> books = new ArrayList<>();
        ArrayList<String> books_names = new ArrayList<>();
        ArrayList<String> books_authors = new ArrayList<>();

        while (scanner.hasNextLine()) {
            String line = scanner.nextLine();
            if (line.isEmpty()) continue;
            int n = Integer.parseInt(line);

            if (n == 1) {
                String line = scanner.nextLine();
                String book = scanner.nextLine();
                String name = scanner.nextLine();
                String author = scanner.nextLine();
                String genre = scanner.nextLine();

                if (line.charAt(0) == 'A' || line.charAt(1) == 'A') {
                    String[] parts = line.split(" ");
                    ArrayList<String> book_parts = new ArrayList<>();
                    for (String part : parts) {
                        if (part.charAt(0) == 'A' || part.charAt(1) == 'A') {
                            book_parts.add(part);
                        }
                    }
                    String title = book_parts.get(0);
                    String author = book_parts.get(1);
                    String genre = book_parts.get(2);

                    Book book = new Book(1, title, author, genre, 0);
                    books.add(book);
                    books_names.add(title);
                    books_authors.add(author);
                }
            }

            if (n == 2) {
                String line = scanner.nextLine();
                if (line.charAt(0) == 'A' || line.charAt(1) == 'A') {
                    Book book = new Book(2, line, line, line, 0);
                    books.add(book);
                    books_names.add(line);
                    books_authors.add(line);
                }
            }

            if (n == 3) {
                String line = scanner.nextLine();
                String title = line;
                String author = line;
                String genre = line;
                Book book = new Book(3, title, author, genre, 0);
                books.add(book);
                books_names.add(title);
                books_authors.add(author);
            }
        }

        System.out.println("Books added successfully.");
    }
}

```

```

        System.out.println(".....");
        for(Book b1){
            System.out.printf("%-25s-%25s\n",b1.t,b1.a,t,b1.au,b1.ct);
        }
    }

    else if(c==4){
        System.out.println();
        System.out.println("Unique Authors");
        for(String au:u){
            System.out.println(au);
        }
    }

    else if(c==5){
        System.out.println();
        System.out.println("Category-wise Book Count");
        LinkedHashMap<String,Integer> count = new LinkedHashMap();
        for(Book b:1){
            count.put(b.ct,count.getOrDefault(b.ct,0)+1);
        }
        for(Map.Entry<String,Integer> e:count.entrySet()){
            System.out.println(e.getKey()+" : "+e.getValue());
        }
    }

    else if(c==6){
        break;
    }
}
sc.close();
}

}

class Book{
    String t,t.au,ct,p;
    Book(String t,String t,String au,String ct,String p){
        this.t=t;
        this.t=t;
        this.au=au;
        this.ct=ct;
        this.p=p;
    }
}
}

```

INPUT

```

1
8081
Java Programming
Herbert Schildt
Programming
Modern Mill
1
8081
Data Structures
Mark Allen
Computer Science
Pearson
4
4
5
2
8081
6

```

OUTPUT

```

(no output)

```